

METAL IMPURITIES IN FOOD ADDITIVES

DO THEY CONTRIBUTE TO DIETARY METAL EXPOSURE?

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Metals in Food Additives?

The **lack of data** on the quantity of metals in the food additives present in the population's diet raises questions. An intake assessment based on EFSA exposure estimates for the Belgian population indicated that, if **Lead (Pb)**, **Cadmium (Cd)** and **Arsenic (As)** are present in food additives (FAs) at concentrations approaching the regulatory limits, the resulting dietary intake would **exceed** current exposure estimate by a **factor of 2 to 12**. Overall metal intake could be **underestimated** if these additives contain more metals than initially assumed.

Our research

To address this data gap, certain food additives were identified as potential sources of metal exposure. Different samples of the selected additives were sampled and analysed for As, Cd and Pb concentrations by inductively coupled plasma mass spectrometry (ICP-MS).



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Relevance



These analyses help to guarantee **food safety** of food products containing additives.



Quantification of metal impurities in selected food additives will provide the data needed to develop a sampling plan for estimating a more accurate **dietary exposure** in the Belgian population.



A comparison between this exposure and previous general exposure estimates will support an **assessment of the risks** related to food additive consumption for the Belgian population.

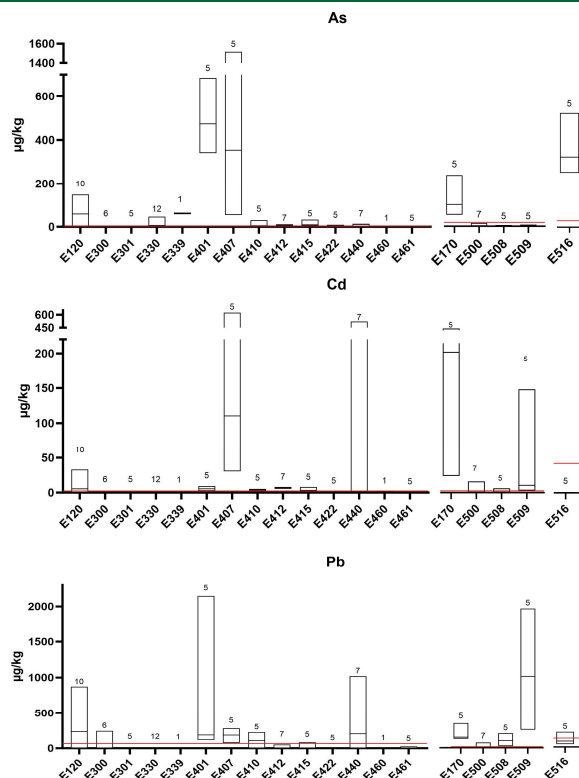


Figure: Metal concentrations (µg/kg) in selected food additives. Box plots show the range and median value. Sample is indicated above the boxes. The red line marks the limit of quantification (LOQ); values below LOQ are included, while those below the limit of detection (LOD = 1/3 LOQ) are set to zero.

Arsenic

Sodium alginate (E401) and **carrageenan** (E407) samples will be analysed further to determine the proportion of inorganic arsenic. The results for **calcium sulphate** (E516) are of particular interest, given the mineral nature of this additive.

Cadmium

Carrageenan (E407) and **calcium carbonate** (E170) have the highest median values. Note that the variability detected for **pectin** (E440) and **calcium chloride** (E509) is attributable to a single sample.

Lead

Notably, the samples with the highest Pb concentrations detected were **sodium alginate** (E401) and **calcium chloride** (E509). Also **carrageenan** (E407) and **pectin** (E440) are important findings due to their high occurrence in food on the Belgian market.

Conclusion

Food additives generally contain low concentrations of metals and all measured values remain **below** the current **regulatory limits**. However, the results show a **wide range** of concentrations both across different food additives and among samples of the same additive sourced from various suppliers. In general, additives of **natural origin** are more likely to have a **higher concentration** of metals. Of all the additives analysed, those with the highest metal concentrations are extracted from plants or animals, in the case of organic additives, or may come from natural extraction, in the case of inorganic additives.



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